Linear Algebra under COVID19

vector add subtract linear equation
inner product
cross product
linear combination span basis
subspace
linear equations augmented matrix
reduced echelon form
affine homogeneous sol. particular sol.
row space column space
Orthogonality ⊥ null space left nullspace ⊥
linearly independence orthogonal basis
orthogonal matrix gram-schmidt

$5x = 10 \qquad x = 2$

$x + 2y = 1$
$3x + 4y = 2$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$A \quad x \quad = \quad b$$

$$\left[\begin{array}{cc|c} 1 & 2 & 1 \\ 3 & 4 & 2 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & 2 & 1 \\ 0 & -2 & -1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & -2 & -1 \end{array}\right] \longrightarrow \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0.5 \end{array}\right]$$

$x = 0, \; y = 0.5$

$x + y = 1$

$$\left[\begin{array}{cc|c} 1 & 1 & 1 \end{array}\right] \qquad \left\{ \begin{pmatrix} -1 \\ 1 \end{pmatrix} y' + \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \; y' \in \mathbb{R} \right\}$$

pivot free in-ome particular

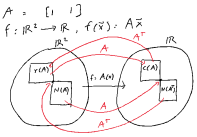
$Ax = b$

$A\vec{x} = \vec{b}$

$f(\vec{x}) = A\vec{x}$

$A = \begin{bmatrix} 1 & 1 \end{bmatrix}$

$f : \mathbb{R}^2 \longrightarrow \mathbb{R}, \; f(\vec{x}) = A\vec{x}$



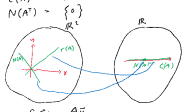
$A = \begin{bmatrix} 1 & 1 \end{bmatrix}$

$r(A) = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\}$

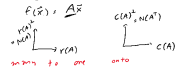
$r(A)^\perp = N(A) = \text{span}\left\{\begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\}$

$c(A) = \mathbb{R}$

$N(A^T) = \{0\}$



$f(\vec{x}) = A\vec{x}$



many to one onto

$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$r(A) = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\} = \mathbb{R}^2$

$c(A) = \text{span}\left\{\begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}\right\} = \mathbb{R}^2$

$N(A) = \left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right\}$

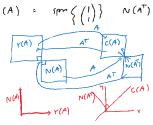
$N(A^T) = \left\{\begin{pmatrix} 0 \\ 0 \end{pmatrix}\right\}$

one to one only

$A = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$

$r(A) = \text{span}\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right\} \qquad N(A) = \text{span}\left\{\begin{pmatrix} 0 \\ 1 \end{pmatrix}\right\}$

$c(A) = \text{span}\left\{\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right\} \qquad N(A^T) = \text{span}\left\{\begin{pmatrix} -1 \\ 1 \end{pmatrix}\right\}$



many to one not onto